

Problem 3

Prove that:

$$1^\circ \quad \overline{\overline{AB}} = A + B; \quad 2^\circ \quad \overline{\overline{C + D}} = CD.$$

Solution

We will prove both identities using De Morgan's laws.

Notation and Definitions

- $\bar{A}$ denotes the complement of A
- AB (or $A \cap B$) denotes the intersection
- $A + B$ (or $A \cup B$) denotes the union

Part 1°: Prove $\overline{\overline{AB}} = A + B$

Step 1: Apply De Morgan's Law

We use De Morgan's law for intersections:

$$\overline{X \cap Y} = \bar{X} \cup \bar{Y}$$

Let $X = \bar{A}$ and $Y = \bar{B}$. Then:

$$\overline{\overline{AB}} = \overline{\overline{A \cap B}} = \overline{\bar{A} \cap \bar{B}} = \bar{\bar{A}} \cup \bar{\bar{B}}$$

Step 2: Simplify Using Double Complement

Since $\bar{\bar{A}} = A$ and $\bar{\bar{B}} = B$:

$$\overline{\overline{AB}} = \bar{\bar{A}} \cup \bar{\bar{B}} = A \cup B = A + B$$

Step 3: Conclusion

Therefore:

$$\boxed{\overline{\overline{AB}} = A + B}$$

Verification Using Elements

An element x is in $\overline{\overline{AB}}$ if and only if:

$$\begin{aligned} x \in \overline{\overline{AB}} &\iff x \notin (\bar{A} \cap \bar{B}) \\ &\iff \text{NOT}(x \in \bar{A} \text{ and } x \in \bar{B}) \\ &\iff x \notin \bar{A} \text{ or } x \notin \bar{B} \\ &\iff x \in A \text{ or } x \in B \\ &\iff x \in A \cup B \\ &\iff x \in A + B \end{aligned}$$

This confirms that $\overline{\overline{AB}} = A + B$.

Part 2°: Prove $\overline{\overline{C} + \overline{D}} = CD$

Step 1: Apply De Morgan's Law

We use De Morgan's law for unions:

$$\overline{X \cup Y} = \bar{X} \cap \bar{Y}$$

Let $X = \bar{C}$ and $Y = \bar{D}$. Then:

$$\overline{\bar{C} + \bar{D}} = \overline{\bar{C} \cup \bar{D}} = \bar{\bar{C}} \cap \bar{\bar{D}}$$

Step 2: Simplify Using Double Complement

Since $\bar{\bar{C}} = C$ and $\bar{\bar{D}} = D$:

$$\bar{\bar{C}} \cap \bar{\bar{D}} = C \cap D = CD$$

Step 3: Conclusion

Therefore:

$$\boxed{\overline{\bar{C} + \bar{D}} = CD}$$

Verification Using Elements

An element x is in $\overline{\bar{C} + \bar{D}}$ if and only if:

$$\begin{aligned} x \in \overline{\bar{C} + \bar{D}} &\iff x \notin (\bar{C} \cup \bar{D}) \\ &\iff x \notin \bar{C} \text{ and } x \notin \bar{D} \\ &\iff x \in C \text{ and } x \in D \\ &\iff x \in C \cap D \\ &\iff x \in CD \end{aligned}$$

This confirms that $\overline{\bar{C} + \bar{D}} = CD$.

De Morgan's Laws Summary

These identities are direct applications of **De Morgan's Laws**:

General Form	Application in This Problem
$\overline{A \cap B} = \bar{A} \cup \bar{B}$	$\overline{\bar{A} \cap \bar{B}} = \bar{\bar{A}} \cup \bar{\bar{B}} = A \cup B$
$\overline{A \cup B} = \bar{A} \cap \bar{B}$	$\overline{\bar{C} \cup \bar{D}} = \bar{\bar{C}} \cap \bar{\bar{D}} = C \cap D$

Intuitive Interpretation

Part 1°: $\overline{\bar{A} \bar{B}} = A + B$

The complement of "neither A nor B " is "at least one of A or B ", which is exactly $A \cup B$.

Part 2°: $\bar{C} + \bar{D} = CD$

The complement of "not C or not D " is "both C and D ", which is exactly $C \cap D$.

Truth Table Verification

For Part 1°

A	B	$\bar{A}$	$\bar{B}$	$\bar{A}\bar{B}$	$\overline{\bar{A}\bar{B}}$	A	B	$A + B$
0	0	1	1	1	0	0	0	0
0	1	1	0	0	1	0	1	1
1	0	0	1	0	1	1	0	1
1	1	0	0	0	1	1	1	1

The columns for $\overline{AB}$ and $A + B$ are identical.

For Part 2°

C	D	$\bar{C}$	$\bar{D}$	$\bar{C} + \bar{D}$	$\overline{\bar{C} + \bar{D}}$	C	D	CD
0	0	1	1	1	0	0	0	0
0	1	1	0	1	0	0	1	0
1	0	0	1	1	0	1	0	0
1	1	0	0	0	1	1	1	1

The columns for $\overline{\bar{C} + \bar{D}}$ and CD are identical.

Final Answer

Part 1°:

$$\boxed{\overline{AB} = \overline{A \cap B} = A \cup B = A + B}$$

Proof: By De Morgan's law, $\overline{X \cap Y} = \bar{X} \cup \bar{Y}$. With $X = \bar{A}$ and $Y = \bar{B}$, we get $\overline{\bar{A} \cap \bar{B}} = \bar{\bar{A}} \cup \bar{\bar{B}} = A \cup B$.

Part 2°:

$$\boxed{\overline{\bar{C} + \bar{D}} = \overline{\bar{C} \cup \bar{D}} = C \cap D = CD}$$

Proof: By De Morgan's law, $\overline{X \cup Y} = \bar{X} \cap \bar{Y}$. With $X = \bar{C}$ and $Y = \bar{D}$, we get $\overline{\bar{C} \cup \bar{D}} = \bar{\bar{C}} \cap \bar{\bar{D}} = C \cap D$.
These are both applications of De Morgan's Laws.